Algoritmo Genético para el TSP

Asignatura: Algoritmos de Optimización

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URL Repositorio GitHub de las 3 Prácticas: *https://github.com/RGS70/03MIAR-Proyecto/tree/main/PR%C3%81CTICAS*
URL GitHub (ipynb): *https://github.com/RGS70/03MIAR-Proyecto/blob/main/PR%C3%81CTICAS/Algoritmo_Genetico_para_el_TSP_Ricardo_Germes_Serrano.ipynb*
URL GitHub (pdf): *https://github.com/RGS70/03MIAR-Proyecto/blob/main/PR%C3%81CTICAS/Algoritmo_Genetico_para_el_TSP_Ricardo_Germes_Serrano.pdf*

Instalación de librerias

In [1]:
```
#Modulo de llamadas http para descargar ficheros
%pip install requests
import urllib.request

#Libreria del problema TSP: http://elib.zib.de/pub/mp-testdata/tsp/tsplib/tsplib.html
%pip install tsplib95
import tsplib95
```

```
Requirement already satisfied: requests in C:\Datos\EV_MASTER\Lib\site-packages (2.34.2)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Datos\EV_MASTER\Lib\site-packages (from requests)
(3.4.7)
Requirement already satisfied: idna<4,>=2.5 in C:\Datos\EV_MASTER\Lib\site-packages (from requests) (3.18)
Requirement already satisfied: urllib3<3,>=1.26 in C:\Datos\EV_MASTER\Lib\site-packages (from requests) (2.7.0)
Requirement already satisfied: certifi>=2023.5.7 in C:\Datos\EV_MASTER\Lib\site-packages (from requests) (2026.5
.20)
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: tsplib95 in C:\Datos\EV_MASTER\Lib\site-packages (0.7.1)
Requirement already satisfied: Click>=6.0 in C:\Datos\EV_MASTER\Lib\site-packages (from tsplib95) (8.4.2)
Requirement already satisfied: Deprecated~=1.2.9 in C:\Datos\EV_MASTER\Lib\site-packages (from tsplib95) (1.2.18
)
Requirement already satisfied: networkx~=2.1 in C:\Datos\EV_MASTER\Lib\site-packages (from tsplib95) (2.8.8)
Requirement already satisfied: tabulate~=0.8.7 in C:\Datos\EV_MASTER\Lib\site-packages (from tsplib95) (0.8.10)
Requirement already satisfied: wrapt<2,>=1.10 in C:\Datos\EV_MASTER\Lib\site-packages (from Deprecated~=1.2.9->t
splib95) (1.17.3)
Requirement already satisfied: colorama in C:\Datos\EV_MASTER\Lib\site-packages (from Click>=6.0->tsplib95) (0.4
.6)
Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
import random   #Libreria para generar numeros y listas aleatorias
import copy     #Permite hacer copias de objetos en python: listas, diccionarios,...
import gzip     #Para descomprimir el fichero .gz sin depender del comando de sistema "gzip"
import shutil
import socket
import os       #Para comprobar si el fichero ya existe en la carpeta local
```

Carga de datos del problema

In [3]:
```
#Librerias y carga del problema

#http://elib.zib.de/pub/mp-testdata/tsp/tsplib/
#Documentacion :
  # https://wwwproxy.iwr.uni-heidelberg.de/groups/comopt/software/TSPLIB95/tsp95.pdf
  # https://tsplib95.readthedocs.io/usage.html
  # https://tsplib95.readthedocs.io/modules.html#module-tsplib95.models

#Limitamos el tiempo de espera de conexión (evita que el notebook se quede
#"colgado" varios minutos si el servidor no responde)
socket.setdefaulttimeout(10)

file = "swiss42.tsp"

#Algunas redes (proxys/firewalls corporativos o de la universidad) no dejan
#conectar con el servidor de la Universidad de Heidelberg. Por eso probamos
#la fuente original y, si falla, comprobamos si el fichero ya está descargado
#en la carpeta local; solo si tampoco está ahí, caemos a un mirror en GitHub
#con los mismos datos (sin necesidad de descomprimir, ya viene en texto plano).
fuente_original = "http://comopt.ifi.uni-heidelberg.de/software/TSPLIB95/tsp/swiss42.tsp.gz"
fuente_alternativa = "https://github.com/mastqe/tsplib/raw/refs/heads/master/swiss42.tsp"
```

```
descargado = False

#1er intento: fuente original (fichero comprimido en .gz)
try:
  urllib.request.urlretrieve(fuente_original, file + '.gz')
  with gzip.open(file + '.gz', 'rb') as f_in:
    with open(file, 'wb') as f_out:
      shutil.copyfileobj(f_in, f_out)
  descargado = True
  print("Datos descargados desde la fuente original (Universidad de Heidelberg).")
except Exception as e:
  print("Aviso: no se ha podido descargar desde la fuente original ->", e)

#2o intento: comprobamos si el fichero ya existe en la carpeta local
if not descargado and os.path.exists(file):
  descargado = True
  print(f"El fichero '{file}' ya existe en la carpeta local. Se usará ese fichero.")

#3er intento (fallback): mirror alternativo en GitHub
if not descargado:
  try:
    urllib.request.urlretrieve(fuente_alternativa, file)
    descargado = True
    print("Datos descargados desde el mirror alternativo de GitHub.")
  except Exception as e:
    print("Aviso: no se ha podido descargar desde el mirror alternativo ->", e)

if not descargado:
  raise RuntimeError("No ha sido posible descargar el fichero swiss42.tsp desde ninguna fuente.")

#Objeto de tsplib95 para nuestro problema problema
problem = tsplib95.load(file)

#Nodos
Nodos = list(problem.get_nodes())

#Aristas
Aristas = list(problem.get_edges())
print("Número de nodos:", len(Nodos))

#Coordenadas(si estan disponibles en el fichero)
problem.get_display(1)

#Distancia
problem.get_weight(1, 2)
```

```
Aviso: no se ha podido descargar desde la fuente original -> <urlopen error timed out>
El fichero 'swiss42.tsp' ya existe en la carpeta local. Se usará ese fichero.
Número de nodos: 42
```

Out[3]: 34

Funciones de la Actividad Guiada 3

In [4]:
```
#Funciones de la Actividad Guiada 3
#Se genera una solucion aleatoria con comienzo en en el nodo 0
def crear_solucion(Nodos):
  solucion = [Nodos[0]]
  for n in Nodos[1:] :
    solucion = solucion + [random.choice(list(set(Nodos) - set(solucion)))]
  return solucion

#Devuelve la distancia entre dos nodos
def distancia(a,b, problem):
  return problem.get_weight(a,b)


#Devuelve la distancia total de una trayectoria/solucion
def distancia_total(solucion, problem):
  distancia_total = 0
  for i in range(len(solucion)-1):
    distancia_total += distancia(solucion[i] ,solucion[i+1] ,  problem)
  return distancia_total + distancia(solucion[len(solucion)-1] ,solucion[0], problem)
```

Funciones Auxiliares

In [5]:
```
#Genera una poblacion inicial de soluciones de tamaño N.
# Puede ser válida la solución aleatoria de la AG3: crear_solucion(Nodos)
def generar_poblacion(Nodos,N):
  poblacion = [crear_solucion(Nodos) for _ in range(N)]
```

```
  return poblacion
```

In [6]:
```
#Evalua la población y devuelve el mejor individuo
def Evaluar_Poblacion(poblacion, problem):
  mejor_solucion = poblacion[0]
  mejor_distancia = distancia_total(poblacion[0], problem)

  for solucion in poblacion[1:]:
    distancia_actual = distancia_total(solucion, problem)
    if distancia_actual < mejor_distancia:
      mejor_solucion = solucion
      mejor_distancia = distancia_actual

  return (mejor_solucion, mejor_distancia)
```

In [7]:
```
#Funcion de cruce. Recibe una poblacion(lista de soluciones) y devuelve la población ampliada con los hijos.
# Todos los individuos de la población son selecionados para el cruce(si la población es par)
# Podría aplicarse un proceso previo de selección para elegir los individuos que se desea cruzar.
def Cruzar(poblacion,mutacion, problem):
  #Barajamos la población para emparejar individuos al azar en cada generación
  #(evita cruzar siempre a los mismos vecinos de la lista)
  poblacion_a_cruzar = copy.deepcopy(poblacion)
  random.shuffle(poblacion_a_cruzar)

  hijos = []
  for i in range(0, len(poblacion_a_cruzar) - 1, 2):
    padres = (poblacion_a_cruzar[i], poblacion_a_cruzar[i+1])
    hijos += Descendencia(padres, problem, mutacion)

  #Devolvemos la población ampliada: padres originales + hijos generados
  return poblacion + hijos
```

In [8]:
```
#Funcion para generar hijos a partir de 2 padres:
# Se elige el metodo de 1-punto de corte pero es posible usar otros n-puntos, uniforme, dependiendo del problema
def Descendencia(padres, problem,mutacion):
  padre1, padre2 = padres
  n = len(padre1)

  #Elegimos un punto de corte (evitamos la posición 0, que es siempre el nodo de inicio)
  punto_corte = random.randint(1, n-1)

  #Hijo 1: primera parte de padre1 + segunda parte de padre2
  hijo1 = padre1[:punto_corte] + padre2[punto_corte:]
  #Hijo 2: primera parte de padre2 + segunda parte de padre1
  hijo2 = padre2[:punto_corte] + padre1[punto_corte:]

  #Los hijos generados así pueden tener nodos repetidos/faltantes: los reparamos
  hijo1 = Factibilizar(hijo1, problem)
  hijo2 = Factibilizar(hijo2, problem)

  #Aplicamos el operador de mutación a cada hijo
  hijo1 = Mutar(hijo1, mutacion)
  hijo2 = Mutar(hijo2, mutacion)

  return [hijo1, hijo2]
```

In [9]:
```
#Para el operador de cruce 1-punto los hijos generados no son soluciones(algunos nodos se repiten y otros no es
def Factibilizar(solucion,problem):
  Nodos = list(problem.get_nodes())

  #Nodos que faltan por visitar en la solución
  faltantes = [n for n in Nodos if n not in solucion]
  random.shuffle(faltantes)

  vistos = set()
  solucion_factible = []
  for nodo in solucion:
    if nodo not in vistos:
      solucion_factible.append(nodo)
      vistos.add(nodo)
    else:
      #Si el nodo ya apareció antes (repetido), lo sustituimos por uno de los que faltan
      nuevo_nodo = faltantes.pop()
      solucion_factible.append(nuevo_nodo)
      vistos.add(nuevo_nodo)

  return solucion_factible
```

In [10]:
```
#Funcion de mutación. Se eligen dos nodos y se intercambia. Se podrian añadir otros operaradores
# Se hace mutaciones mutacion% de las veces
def Mutar(solucion,mutacion):
  if random.random() <= mutacion:
```

```
    #Elegimos dos posiciones al azar (evitamos la posición 0, nodo de inicio fijo)
    i, j = random.sample(range(1, len(solucion)), 2)
    solucion[i], solucion[j] = solucion[j], solucion[i]
  return solucion
```

In [11]:
```
#Funcion de seleccion de la población. Recibe como parametro una poblacion y
# devuelve una poblacion a la que se ha eliminado individuos poco aptos(fitness alto) y para mantener una pobla
#Se tiene en cuenta el porcentaje elitismo pasado como parametro
# Para los individuos que no son de la elite podríamos usar una selección de ruleta(proporcional a su fitness)
def Seleccionar(problem,poblacion, N, elitismo):
  #Ordenamos la población de mejor a peor (menor distancia = mejor)
  poblacion_ordenada = sorted(poblacion, key=lambda solucion: distancia_total(solucion, problem))

  #Nos quedamos siempre con la élite (los mejores individuos)
  n_elite = int(N * elitismo)
  elite = poblacion_ordenada[:n_elite]

  #El resto hasta completar N se elige mediante selección de ruleta, proporcional al fitness
  #(a menor distancia, mayor fitness = mayor probabilidad de ser seleccionado)
  resto_poblacion = poblacion_ordenada[n_elite:]
  distancias = [distancia_total(s, problem) for s in resto_poblacion]
  fitness = [1 / d for d in distancias]
  total_fitness = sum(fitness)
  probabilidades = [f / total_fitness for f in fitness]

  n_restantes = N - n_elite
  seleccionados = random.choices(resto_poblacion, weights=probabilidades, k=n_restantes) if resto_poblacion else

  return elite + seleccionados
```

Proceso Principal

In [12]:
```
#Funcion principal del algoritmo genetico
##########################################################3
def algoritmo_genetico(problem=problem,N=100,mutacion=.15,elitismo=.1,generaciones=100):
  # problem = datos del problema
  # N = Tamaño de la población
  # mutacion = probabilidad de una mutación
  # elitismo = porcion de la mejor poblacion a mantener
  # generaciones = nº de generaciones a generar para finalizar

  #Genera la poblacion inicial
  Nodos = list(problem.get_nodes())
  poblacion = generar_poblacion(Nodos,N)

  #Inicializamos valores para la mejor solucion
  (mejor_solucion, mejor_distancia) = Evaluar_Poblacion(poblacion, problem)


  #Condicion de parada
  parar = False
  n=0
  #Inciamos el cliclo de generaciones
  while(parar == False) :

    #Cruce de la poblacion(incluye mutación)
    poblacion = Cruzar(poblacion, mutacion, problem)

    #Seleccionamos la población
    poblacion = Seleccionar(problem,poblacion, N, elitismo)

    #Evaluamos la nueva población
    (mejor_solucion, mejor_distancia) = Evaluar_Poblacion(poblacion, problem)

    print("Generacion #", n, "\nLa mejor solución es:" , mejor_solucion, "\ncon distancia " , mejor_distancia,

    #Numero de generaciones. Criterio de parada
    if n==generaciones:
      parar = True
    n +=1

  return mejor_solucion


sol = algoritmo_genetico(problem=problem,N=500,mutacion=.3,elitismo=.40,generaciones=250)
```

```
Generacion # 0
La mejor solución es: [0, 23, 8, 22, 10, 3, 4, 38, 9, 41, 35, 32, 1, 25, 26, 31, 5, 19, 34, 28, 2, 27, 12, 6, 14
, 36, 18, 13, 30, 29, 33, 17, 7, 21, 20, 39, 24, 40, 11, 15, 37, 16]
con distancia  3895
```

```
Generacion # 1
La mejor solución es: [0, 23, 8, 22, 10, 3, 4, 38, 9, 41, 35, 32, 1, 25, 26, 31, 5, 19, 34, 28, 2, 27, 12, 6, 14
, 36, 18, 13, 30, 29, 33, 17, 7, 21, 20, 39, 24, 40, 11, 15, 37, 16]
con distancia  3895

Generacion # 2
La mejor solución es: [0, 35, 24, 40, 9, 8, 2, 3, 32, 7, 39, 20, 27, 11, 22, 5, 15, 17, 36, 31, 38, 25, 26, 12,
10, 30, 21, 34, 1, 4, 18, 29, 28, 41, 23, 19, 33, 37, 14, 13, 6, 16]
con distancia  3827

Generacion # 3
La mejor solución es: [0, 35, 24, 40, 9, 8, 2, 3, 32, 7, 39, 20, 27, 11, 22, 5, 15, 17, 36, 31, 38, 25, 26, 12,
10, 30, 21, 34, 1, 4, 18, 29, 28, 41, 23, 19, 33, 37, 14, 13, 6, 16]
con distancia  3827

Generacion # 4
La mejor solución es: [0, 35, 14, 6, 9, 24, 40, 34, 33, 41, 18, 26, 20, 31, 5, 16, 12, 25, 11, 13, 28, 10, 7, 2,
15, 37, 32, 1, 22, 8, 39, 21, 38, 23, 29, 3, 27, 19, 36, 17, 30, 4]
con distancia  3559

Generacion # 5
La mejor solución es: [0, 35, 14, 6, 9, 24, 40, 34, 33, 41, 18, 26, 20, 31, 5, 16, 12, 25, 11, 13, 28, 10, 7, 2,
15, 37, 32, 1, 22, 8, 39, 21, 38, 23, 29, 3, 27, 19, 36, 17, 30, 4]
con distancia  3559

Generacion # 6
La mejor solución es: [0, 35, 14, 6, 9, 24, 40, 34, 33, 41, 18, 26, 20, 31, 5, 16, 12, 25, 11, 13, 28, 10, 7, 2,
15, 37, 32, 4, 22, 8, 39, 21, 38, 23, 29, 3, 27, 19, 36, 17, 30, 1]
con distancia  3544

Generacion # 7
La mejor solución es: [0, 26, 22, 11, 12, 41, 13, 19, 16, 15, 36, 32, 38, 39, 7, 37, 6, 35, 17, 14, 9, 5, 4, 1,
2, 8, 25, 18, 10, 27, 23, 33, 24, 21, 40, 31, 3, 28, 20, 34, 30, 29]
con distancia  3527

Generacion # 8
La mejor solución es: [0, 23, 8, 22, 10, 3, 4, 38, 9, 41, 5, 32, 1, 25, 26, 31, 19, 28, 2, 29, 6, 33, 30, 24, 21
, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 40, 15, 17, 7, 16, 14]
con distancia  3490

Generacion # 9
La mejor solución es: [0, 23, 8, 22, 10, 3, 4, 38, 9, 41, 5, 32, 1, 25, 26, 31, 19, 28, 2, 29, 6, 33, 30, 24, 21
, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 40, 15, 17, 7, 16, 14]
con distancia  3490

Generacion # 10
La mejor solución es: [0, 23, 8, 22, 10, 3, 4, 38, 9, 41, 5, 32, 1, 25, 26, 31, 19, 28, 2, 29, 6, 33, 30, 24, 21
, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 40, 15, 17, 7, 16, 14]
con distancia  3490

Generacion # 11
La mejor solución es: [0, 23, 8, 22, 10, 3, 4, 38, 9, 41, 5, 32, 1, 25, 26, 31, 19, 28, 2, 29, 6, 33, 30, 24, 21
, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 40, 15, 17, 7, 16, 14]
con distancia  3490

Generacion # 12
La mejor solución es: [0, 26, 22, 11, 12, 41, 13, 19, 16, 15, 36, 32, 38, 39, 7, 37, 20, 35, 17, 14, 9, 5, 4, 1,
2, 8, 25, 18, 10, 33, 24, 30, 23, 21, 40, 28, 34, 31, 6, 29, 27, 3]
con distancia  3458

Generacion # 13
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 29, 3, 41, 11, 7, 22, 30, 4, 5, 1, 21, 9, 8, 1
8, 26, 34, 10, 12, 23, 24, 40, 39, 32, 27, 2, 6, 38, 25, 17, 28, 33]
con distancia  3440

Generacion # 14
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 29, 3, 41, 11, 7, 22, 30, 4, 5, 1, 21, 9, 8, 1
8, 26, 34, 10, 12, 23, 24, 40, 39, 32, 27, 2, 6, 38, 25, 17, 28, 33]
con distancia  3440

Generacion # 15
La mejor solución es: [0, 23, 8, 22, 30, 7, 6, 33, 20, 34, 35, 28, 27, 11, 19, 5, 15, 17, 36, 31, 38, 2, 25, 12,
10, 9, 21, 24, 40, 37, 13, 41, 3, 32, 39, 1, 18, 14, 16, 4, 26, 29]
con distancia  3358

Generacion # 16
La mejor solución es: [0, 23, 8, 22, 30, 3, 4, 38, 9, 41, 5, 32, 40, 25, 26, 31, 19, 28, 2, 29, 6, 33, 10, 24, 2
1, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 7, 14, 16, 15, 17, 1]
con distancia  3244

Generacion # 17
La mejor solución es: [0, 26, 22, 8, 30, 3, 4, 6, 9, 41, 5, 32, 40, 25, 23, 31, 19, 28, 2, 29, 38, 33, 10, 24, 2
1, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 7, 14, 16, 15, 17, 1]
```

```
con distancia  3208

Generacion # 18
La mejor solución es: [0, 26, 22, 8, 30, 3, 4, 6, 9, 41, 5, 32, 40, 25, 23, 31, 19, 28, 2, 29, 38, 33, 10, 24, 2
1, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 7, 14, 16, 15, 17, 1]
con distancia  3208

Generacion # 19
La mejor solución es: [0, 26, 22, 8, 30, 3, 4, 6, 9, 41, 5, 32, 40, 25, 23, 31, 19, 28, 2, 29, 38, 33, 10, 24, 2
1, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 7, 14, 16, 15, 17, 1]
con distancia  3208

Generacion # 20
La mejor solución es: [0, 26, 22, 8, 30, 3, 4, 6, 9, 41, 5, 19, 40, 25, 23, 31, 32, 28, 2, 29, 38, 33, 10, 24, 2
1, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 7, 15, 14, 17, 16, 1]
con distancia  3164

Generacion # 21
La mejor solución es: [0, 26, 22, 8, 30, 3, 4, 6, 9, 41, 5, 19, 40, 25, 23, 31, 32, 28, 2, 29, 38, 33, 10, 24, 2
1, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 7, 15, 14, 17, 16, 1]
con distancia  3164

Generacion # 22
La mejor solución es: [0, 26, 22, 8, 30, 3, 4, 6, 9, 41, 5, 19, 40, 25, 23, 31, 32, 28, 2, 29, 38, 33, 10, 24, 2
1, 39, 27, 11, 18, 12, 13, 37, 35, 36, 34, 20, 7, 15, 14, 17, 16, 1]
con distancia  3164

Generacion # 23
La mejor solución es: [0, 23, 8, 22, 30, 7, 6, 33, 20, 34, 35, 28, 27, 11, 19, 5, 15, 17, 36, 31, 38, 2, 25, 12,
10, 9, 21, 24, 40, 37, 13, 18, 3, 32, 4, 29, 39, 41, 26, 14, 16, 1]
con distancia  3046

Generacion # 24
La mejor solución es: [0, 23, 8, 22, 30, 7, 6, 33, 20, 34, 35, 28, 27, 11, 19, 5, 15, 17, 36, 31, 38, 2, 25, 12,
10, 9, 21, 24, 40, 37, 13, 18, 3, 32, 4, 29, 39, 41, 26, 14, 16, 1]
con distancia  3046

Generacion # 25
La mejor solución es: [0, 23, 8, 22, 30, 7, 6, 33, 20, 34, 35, 28, 27, 11, 19, 5, 15, 17, 36, 31, 38, 2, 25, 12,
10, 9, 21, 24, 40, 37, 13, 18, 3, 32, 4, 29, 39, 41, 26, 14, 16, 1]
con distancia  3046

Generacion # 26
La mejor solución es: [0, 29, 22, 11, 12, 20, 13, 19, 16, 15, 36, 32, 38, 39, 7, 37, 6, 35, 17, 14, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 30, 9, 34, 31, 33, 28, 3, 27]
con distancia  3034

Generacion # 27
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 32, 38, 39, 7, 12, 11, 41, 17, 26, 5, 3, 1, 4,
2, 8, 27, 9, 21, 40, 24, 30, 23, 29, 22, 6, 33, 34, 28, 10, 25, 18]
con distancia  2997

Generacion # 28
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 32, 38, 39, 7, 12, 11, 41, 17, 26, 5, 3, 1, 4,
2, 8, 27, 9, 21, 40, 24, 30, 23, 29, 22, 6, 33, 34, 28, 10, 25, 18]
con distancia  2997

Generacion # 29
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 27, 11, 26, 5, 15, 17, 36, 33, 38, 2, 4, 1
, 6, 12, 9, 23, 21, 40, 24, 32, 7, 18, 10, 39, 25, 29, 41, 8, 3, 30]
con distancia  2890

Generacion # 30
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 27, 11, 26, 5, 15, 17, 36, 33, 38, 2, 4, 1
, 6, 12, 9, 23, 21, 40, 24, 32, 7, 18, 10, 39, 25, 29, 41, 8, 3, 30]
con distancia  2890

Generacion # 31
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 27, 11, 26, 5, 15, 17, 36, 33, 38, 2, 4, 1
, 6, 12, 9, 23, 21, 40, 24, 32, 7, 18, 10, 39, 25, 29, 41, 8, 3, 30]
con distancia  2890

Generacion # 32
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 27, 11, 26, 5, 15, 17, 36, 33, 38, 2, 4, 1
, 6, 12, 9, 23, 21, 40, 24, 32, 7, 18, 10, 39, 25, 29, 41, 8, 3, 30]
con distancia  2890

Generacion # 33
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 27, 11, 26, 5, 15, 17, 36, 33, 38, 2, 4, 1
, 6, 12, 9, 23, 21, 40, 24, 32, 7, 18, 10, 39, 25, 29, 41, 8, 3, 30]
con distancia  2890

Generacion # 34
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La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 27, 11, 26, 5, 15, 17, 36, 33, 38, 2, 4, 1
, 6, 12, 9, 23, 21, 40, 24, 32, 7, 18, 10, 39, 25, 29, 41, 8, 30, 3]
con distancia  2839

Generacion # 35
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 32, 11, 26, 5, 15, 17, 36, 33, 38, 2, 4, 1
, 6, 12, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 7, 29, 8, 30, 3, 27]
con distancia  2733

Generacion # 36
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 32, 11, 26, 5, 15, 17, 36, 33, 38, 2, 4, 1
, 6, 12, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 7, 29, 8, 30, 3, 27]
con distancia  2733

Generacion # 37
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 27, 11, 26, 5, 15, 17, 36, 33, 38, 2, 25,
12, 10, 9, 21, 24, 40, 23, 8, 18, 3, 30, 4, 29, 39, 41, 6, 7, 1, 32]
con distancia  2710

Generacion # 38
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 34, 22, 28, 27, 11, 26, 5, 15, 17, 36, 33, 38, 2, 25,
12, 10, 9, 21, 24, 40, 23, 8, 18, 3, 30, 4, 29, 39, 41, 6, 7, 1, 32]
con distancia  2710

Generacion # 39
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 27, 30]
con distancia  2644

Generacion # 40
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 27, 30]
con distancia  2644

Generacion # 41
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 27, 30]
con distancia  2644

Generacion # 42
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 30, 27]
con distancia  2616

Generacion # 43
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 30, 27]
con distancia  2616

Generacion # 44
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 30, 27]
con distancia  2616

Generacion # 45
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 30, 27]
con distancia  2616

Generacion # 46
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 30, 27]
con distancia  2616

Generacion # 47
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 30, 27]
con distancia  2616

Generacion # 48
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 30, 27]
con distancia  2616

Generacion # 49
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 39, 29, 12, 6, 32, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 28, 30, 27]
con distancia  2616

Generacion # 50
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 32, 38, 39, 12, 11, 5, 17, 34, 33, 28, 2, 25,
8, 10, 9, 21, 24, 40, 22, 41, 23, 18, 6, 1, 7, 26, 29, 4, 30, 27, 3]
con distancia  2583
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Generacion # 51
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 32, 38, 39, 12, 11, 5, 17, 34, 33, 28, 2, 25,
8, 10, 9, 21, 24, 40, 22, 41, 23, 18, 6, 1, 7, 26, 29, 4, 30, 27, 3]
con distancia  2583

Generacion # 52
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 32, 38, 39, 12, 11, 5, 17, 34, 33, 28, 2, 25,
8, 10, 9, 21, 24, 40, 22, 41, 23, 18, 6, 1, 7, 26, 29, 4, 30, 27, 3]
con distancia  2583

Generacion # 53
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 32, 38, 39, 12, 11, 5, 17, 34, 33, 28, 2, 25,
8, 10, 9, 21, 24, 40, 22, 41, 23, 18, 6, 1, 7, 26, 29, 4, 30, 27, 3]
con distancia  2583

Generacion # 54
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 32, 39, 29, 12, 6, 3, 17, 11, 26, 5, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 34, 33, 38, 22, 30, 28, 27]
con distancia  2544

Generacion # 55
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 28, 33, 32, 2, 4, 1,
6, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 34, 38, 29, 30, 27]
con distancia  2440

Generacion # 56
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 28, 33, 32, 2, 4, 1,
6, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 34, 38, 29, 30, 27]
con distancia  2440

Generacion # 57
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 28, 33, 32, 2, 4, 1,
6, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 34, 38, 29, 30, 27]
con distancia  2440

Generacion # 58
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 28, 33, 32, 2, 4, 1,
6, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 34, 38, 29, 30, 27]
con distancia  2440

Generacion # 59
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 22, 38, 18, 10, 25, 41, 39, 28, 8, 29, 30]
con distancia  2373

Generacion # 60
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 22, 38, 18, 10, 25, 41, 39, 28, 8, 29, 30]
con distancia  2373

Generacion # 61
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 22, 38, 18, 10, 25, 41, 39, 28, 8, 29, 30]
con distancia  2373

Generacion # 62
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 22, 38, 18, 10, 25, 41, 39, 28, 8, 29, 30]
con distancia  2373

Generacion # 63
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 22, 38, 18, 10, 25, 41, 39, 28, 8, 29, 30]
con distancia  2373

Generacion # 64
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 29, 22, 30, 38, 28]
con distancia  2323

Generacion # 65
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 29, 22, 30, 38, 28]
con distancia  2323

Generacion # 66
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 29, 22, 30, 38, 28]
con distancia  2323

Generacion # 67
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
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6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 29, 22, 30, 38, 28]
con distancia 2323

Generacion # 68
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 29, 22, 30, 38, 28]
con distancia 2323

Generacion # 69
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 29, 22, 30, 38, 28]
con distancia 2323

Generacion # 70
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 6, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 38, 28, 30, 29, 27]
con distancia 2249

Generacion # 71
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 6, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 38, 28, 30, 29, 27]
con distancia 2249

Generacion # 72
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 6, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 38, 28, 30, 29, 27]
con distancia 2249

Generacion # 73
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 6, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 38, 28, 30, 29, 27]
con distancia 2249

Generacion # 74
La mejor solución es: [0, 35, 20, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 6, 4, 1,
2, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 38, 28, 30, 29, 27]
con distancia 2249

Generacion # 75
La mejor solución es: [0, 20, 35, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 29, 28, 30, 38, 22]
con distancia 2236

Generacion # 76
La mejor solución es: [0, 20, 35, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 30, 38, 22, 29, 28]
con distancia 2211

Generacion # 77
La mejor solución es: [0, 35, 17, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 8, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 29, 28, 27, 30, 22, 38]
con distancia 2204

Generacion # 78
La mejor solución es: [0, 35, 17, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 8, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 29, 28, 27, 30, 22, 38]
con distancia 2204

Generacion # 79
La mejor solución es: [0, 20, 35, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 38, 28, 30, 29, 27]
con distancia 2187

Generacion # 80
La mejor solución es: [0, 20, 35, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 8, 25, 18, 10, 41, 40, 21, 24, 23, 9, 39, 22, 38, 28, 30, 29, 27]
con distancia 2187

Generacion # 81
La mejor solución es: [0, 35, 17, 31, 37, 14, 13, 19, 16, 15, 36, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 8, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 29, 28, 30, 38, 22, 27]
con distancia 2180

Generacion # 82
La mejor solución es: [0, 20, 35, 31, 36, 14, 13, 19, 16, 15, 37, 7, 3, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 29, 22, 38, 30, 28]
con distancia 2135

Generacion # 83
La mejor solución es: [0, 35, 36, 31, 37, 14, 13, 19, 16, 15, 17, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 8, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 27, 29, 30, 38, 22, 28]
con distancia 2112

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Generacion # 84
La mejor solución es: [0, 35, 36, 31, 37, 14, 13, 19, 16, 15, 17, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 8, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 27, 29, 30, 38, 22, 28]
con distancia  2112

Generacion # 85
La mejor solución es: [0, 20, 35, 31, 36, 14, 13, 19, 16, 15, 37, 7, 6, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
3, 28, 18, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 22, 38, 29, 30, 27]
con distancia  2101

Generacion # 86
La mejor solución es: [0, 20, 35, 31, 36, 14, 13, 19, 16, 15, 37, 7, 6, 11, 12, 5, 26, 17, 34, 33, 32, 2, 4, 1,
3, 28, 18, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 22, 38, 29, 30, 27]
con distancia  2101

Generacion # 87
La mejor solución es: [0, 35, 36, 31, 37, 14, 13, 19, 16, 15, 17, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 27, 9, 23, 21, 40, 24, 39, 18, 10, 25, 41, 8, 29, 30, 38, 22, 28]
con distancia  2057

Generacion # 88
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 6, 4, 1,
2, 28, 18, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 22, 38, 29, 30, 27]
con distancia  2010

Generacion # 89
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 6, 4, 1,
2, 28, 18, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 22, 38, 29, 30, 27]
con distancia  2010

Generacion # 90
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 6, 4, 1,
2, 28, 18, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 22, 38, 29, 30, 27]
con distancia  2010

Generacion # 91
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 28, 18, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 22, 38, 29, 30, 27]
con distancia  2006

Generacion # 92
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 3, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 28, 18, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 29, 22, 38, 30, 27]
con distancia  1993

Generacion # 93
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 28, 3, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 29, 22, 38, 30, 27]
con distancia  1948

Generacion # 94
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 28, 3, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 29, 22, 38, 30, 27]
con distancia  1948

Generacion # 95
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 28, 3, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 29, 22, 38, 30, 27]
con distancia  1948

Generacion # 96
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 28, 3, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 29, 22, 38, 30, 27]
con distancia  1948

Generacion # 97
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 28, 3, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 29, 22, 38, 30, 27]
con distancia  1948

Generacion # 98
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 6, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
3, 27, 18, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1915

Generacion # 99
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 1, 11, 12, 5, 28, 20, 34, 33, 32, 2, 4, 3,
6, 26, 18, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 29, 22, 38, 30, 27]
con distancia  1913

Generacion # 100
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 27, 3, 23, 21, 40, 24, 39, 9, 10, 25, 41, 8, 29, 22, 38, 30, 28]
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con distancia  1910

Generacion # 101
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 27, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1900

Generacion # 102
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 27, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1900

Generacion # 103
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 27, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1900

Generacion # 104
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 1,
6, 27, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1900

Generacion # 105
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 4, 6,
1, 27, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1889

Generacion # 106
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 11, 12, 5, 7, 20, 34, 33, 32, 2, 4, 3,
6, 1, 28, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 27]
con distancia  1887

Generacion # 107
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1875

Generacion # 108
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1875

Generacion # 109
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 11, 12, 5, 7, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1862

Generacion # 110
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 11, 12, 5, 7, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1862

Generacion # 111
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 11, 12, 5, 7, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1862

Generacion # 112
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 11, 12, 5, 7, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1862

Generacion # 113
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 11, 12, 5, 7, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1862

Generacion # 114
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 11, 12, 5, 7, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1862

Generacion # 115
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1854

Generacion # 116
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1854

Generacion # 117
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La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1854

Generacion # 118
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 12, 11, 5, 7, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1851

Generacion # 119
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 12, 11, 5, 7, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1851

Generacion # 120
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 12, 11, 5, 7, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1851

Generacion # 121
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 26, 5, 20, 34, 33, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1848

Generacion # 122
La mejor solución es: [0, 31, 36, 35, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1834

Generacion # 123
La mejor solución es: [0, 31, 36, 35, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1834

Generacion # 124
La mejor solución es: [0, 31, 36, 35, 17, 14, 13, 19, 16, 15, 37, 7, 18, 11, 12, 5, 26, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1834

Generacion # 125
La mejor solución es: [0, 35, 36, 31, 17, 14, 13, 19, 16, 15, 37, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 3, 4, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1831

Generacion # 126
La mejor solución es: [0, 35, 36, 31, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1810

Generacion # 127
La mejor solución es: [0, 35, 36, 31, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 8, 10, 25, 41, 9, 22, 38, 30, 29, 28]
con distancia  1810

Generacion # 128
La mejor solución es: [0, 35, 36, 31, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1799

Generacion # 129
La mejor solución es: [0, 35, 36, 31, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1799

Generacion # 130
La mejor solución es: [0, 35, 36, 31, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1799

Generacion # 131
La mejor solución es: [0, 35, 36, 31, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1799

Generacion # 132
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1790

Generacion # 133
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1790
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Generacion # 134
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1790

Generacion # 135
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1790

Generacion # 136
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1790

Generacion # 137
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1789

Generacion # 138
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1789

Generacion # 139
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1789

Generacion # 140
La mejor solución es: [0, 35, 36, 31, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1776

Generacion # 141
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 19, 16, 15, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 23, 21, 40, 24, 39, 9, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1775

Generacion # 142
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1767

Generacion # 143
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1766

Generacion # 144
La mejor solución es: [0, 31, 36, 35, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1766

Generacion # 145
La mejor solución es: [0, 31, 35, 36, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1754

Generacion # 146
La mejor solución es: [0, 31, 35, 36, 17, 37, 13, 19, 16, 15, 14, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1754

Generacion # 147
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 19, 16, 15, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia  1752

Generacion # 148
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 19, 16, 15, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1751

Generacion # 149
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 19, 16, 15, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1751

Generacion # 150
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 19, 16, 15, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
```

, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1751

Generacion # 151
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 19, 16, 15, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia 1739

Generacion # 152
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 19, 16, 15, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia 1739

Generacion # 153
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 19, 16, 15, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia 1739

Generacion # 154
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia 1682

Generacion # 155
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia 1682

Generacion # 156
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia 1682

Generacion # 157
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1681

Generacion # 158
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1681

Generacion # 159
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1681

Generacion # 160
La mejor solución es: [0, 31, 36, 35, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1681

Generacion # 161
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 22, 38, 30, 29, 28]
con distancia 1669

Generacion # 162
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1668

Generacion # 163
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1668

Generacion # 164
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1668

Generacion # 165
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1668

Generacion # 166
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia 1668

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Generacion # 167
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1668

Generacion # 168
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 29, 21, 40, 24, 39, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1661

Generacion # 169
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 29, 21, 40, 24, 39, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1661

Generacion # 170
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 29, 21, 40, 24, 39, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1661

Generacion # 171
La mejor solución es: [0, 31, 35, 36, 17, 37, 14, 15, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 29, 21, 40, 24, 39, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1661

Generacion # 172
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1653

Generacion # 173
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1653

Generacion # 174
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1653

Generacion # 175
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1653

Generacion # 176
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1653

Generacion # 177
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 9, 21, 40, 24, 39, 23, 41, 25, 10, 8, 29, 22, 38, 30, 28]
con distancia  1653

Generacion # 178
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 29, 21, 40, 24, 39, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1648

Generacion # 179
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 21, 40, 24, 39, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1643

Generacion # 180
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 21, 40, 24, 39, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1643

Generacion # 181
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 21, 40, 24, 39, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1643

Generacion # 182
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 40, 24, 21, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1629

Generacion # 183
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 40, 24, 21, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
```

```
con distancia  1629

Generacion # 184
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 40, 24, 21, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1629

Generacion # 185
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 40, 24, 21, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1629

Generacion # 186
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 40, 24, 21, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1629

Generacion # 187
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 40, 24, 21, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1629

Generacion # 188
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 40, 24, 21, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1629

Generacion # 189
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 40, 24, 21, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1629

Generacion # 190
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1612

Generacion # 191
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1612

Generacion # 192
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1612

Generacion # 193
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 27, 1
, 6, 4, 3, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1612

Generacion # 194
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 2, 1
, 6, 4, 3, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1610

Generacion # 195
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 2, 1
, 6, 4, 3, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1610

Generacion # 196
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1607

Generacion # 197
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1607

Generacion # 198
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 2, 3, 1,
6, 4, 27, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1607

Generacion # 199
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 200
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La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 201
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 202
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 203
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 204
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 205
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 206
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 207
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 208
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 14, 16, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1602

Generacion # 209
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 210
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 211
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 212
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 213
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 214
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 215
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 216
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598
```

```
Generacion # 217
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 218
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 219
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 220
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 221
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 222
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 223
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 224
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 225
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 226
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 227
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 228
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 229
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 230
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 231
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 232
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598

Generacion # 233
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
```

, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 234
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 235
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 236
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 237
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 238
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 239
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 240
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 241
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 242
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 243
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 244
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 245
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 246
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 247
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 248
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

Generacion # 249
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia 1598

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Generacion # 250
La mejor solución es: [0, 31, 35, 36, 17, 37, 15, 16, 14, 19, 13, 26, 18, 11, 12, 5, 7, 20, 33, 34, 32, 27, 3, 1
, 6, 4, 2, 29, 39, 21, 24, 40, 23, 41, 25, 10, 8, 9, 22, 38, 30, 28]
con distancia  1598
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In [ ]: